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**A MINOR PROJECT PROPOSAL ON
DECENTRALIZED PREPAID SMART POWER GRID**

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Abstract

The increasing demand for electricity and the need for efficient energy management have created a need for smarter and more reliable power distribution systems. Traditional electricity metering and billing systems often depend on centralized infrastructure, manual meter reading, and postpaid billing, which can lead to delayed payments, energy wastage, and limited monitoring of electricity consumption. This project proposes a **Decentralized Prepaid Smart Power Grid** that combines smart energy metering, prepaid electricity management, real-time monitoring, and decentralized control to improve the efficiency and reliability of electricity distribution.

The proposed system enables consumers to monitor their electricity consumption and available energy credit in real time. Electricity usage is measured using smart metering technology, while a prepaid mechanism allows consumers to purchase and manage electricity credit before consumption. The system is designed to provide automatic monitoring of energy usage and balance, with the ability to disconnect or restrict power supply when the available credit is exhausted. A decentralized architecture is proposed to reduce dependency on a single centralized system and improve system reliability, scalability, and transparency.

The project aims to develop a practical and cost-effective prototype that demonstrates efficient electricity consumption monitoring, prepaid energy management, and decentralized power control. The proposed system is expected to reduce manual intervention, improve energy awareness among consumers, minimize billing-related issues, and provide a more flexible approach to future smart-grid applications.

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LIST OF ABBREVIATIONS

API : Application Programming Interface

Colab : Colaboratory

CHAPTER 1

INTRODUCTION

1.1 Background

Hello World

1.2 Gap Identification

1.3 Motivation

1.4 Objectives

- Primary objective goes here
- Secondary objective goes here

CHAPTER 2

RELATED THEORY

2.1 Related Theory goes here

CHAPTER 3

LITERATURE REVIEW

Literature review goes here

CHAPTER 4

METHODOLOGY

4.1 Overview

4.2 Other section goes here

CHAPTER 5

EXPECTED RESULTS

REFERENCES

APPENDIX A

APPENDIX B